

Java Collections Cheat Sheet

ArrayList

Method	Description	Example
add(E)	Add element at end	list.add("Apple");
add(int, E)	Insert at index	list.add(1, "Banana");
get(int)	Get element	list.get(0);
set(int, E)	Replace element	list.set(0, "Cherry");
remove(int) / remove(Object)	Remove by index or value	list.remove("Apple");
contains(Object)	Check existence	list.contains("Banana");
size() / isEmpty()	Count / empty check	list.size();
indexOf() / lastIndexOf()	Search index	list.indexOf("Apple");
clear()	Remove all	list.clear();
addAll() / removeAll() / retainAll()	Bulk operations	list.addAll(other);
toArray() / toArray(T[])	Convert to array	list.toArray(new String[0]);
clone()	Shallow copy	(ArrayList<String>) list.clone();
ensureCapacity() / trimToSize()	Capacity tuning	list.ensureCapacity(20);
removeIf()	Conditional removal	list.removeIf(s -> s.startsWith("A"));
replaceAll()	Apply function	list.replaceAll(String::toUpperCase);
sort()	Sort list	list.sort(String::compareTo);
subList()	View portion	list.subList(0, 2);

LinkedList

Method	Description	Example
addFirst() / addLast()	Add at head/tail	list.addFirst("Apple");
removeFirst() / removeLast()	Remove head/tail	list.removeFirst();
getFirst() / getLast()	Get head/tail	list.getFirst();
offer() / offerFirst() / offerLast()	Queue-style add	list.offer("Banana");
poll() / pollFirst() / pollLast()	Queue-style remove	list.poll();
peek() / peekFirst() / peekLast()	Queue-style peek	list.peek();
push() / pop()	Stack operations	list.push("Cherry"); list.pop();
All ArrayList methods	Inherited	list.get(0);

HashSet

Method	Description	Example
add(E)	Add element	set.add("Apple");
remove(Object)	Remove element	set.remove("Apple");
contains(Object)	Check existence	set.contains("Banana");
size() / isEmpty()	Count / empty check	set.size();
clear()	Remove all	set.clear();
addAll() / removeAll() / retainAll()	Bulk operations	set.addAll(otherSet);
iterator()	Iterate	for(String s : set) {}

TreeSet

Method	Description	Example
first() / last()	Get smallest/largest	set.first();
headSet(E) / tailSet(E)	Range views	set.headSet("Mango");
subSet(E, E)	Range subset	set.subSet("A", "Z");
higher(E) / lower(E)	Next higher/lower	set.higher("Apple");
ceiling(E) / floor(E)	>= or <= element	set.ceiling("Banana");
All HashSet methods	Inherited	set.add("Apple");

LinkedHashSet

Method	Description	Example
Preserves insertion order	Iteration order fixed	set.add("Apple");
All HashSet methods	Same as HashSet	set.contains("Banana");

HashMap

Method	Description	Example
put(K,V)	Insert key-value	map.put(1, "Apple");
get(K)	Retrieve value	map.get(1);
remove(K)	Remove entry	map.remove(1);
containsKey() / containsValue()	Existence check	map.containsKey(2);
size() / isEmpty()	Count / empty check	map.size();
replace()	Update value	map.replace(2, "Banana");
putIfAbsent()	Conditional insert	map.putIfAbsent(3, "Cherry");
keySet() / values() / entrySet()	Views	map.keySet();
compute() / computeIfAbsent() / computeIfPresent()	Full conditional updates	map.compute(1, (k,v)->"X");

merge()	Merge values	map.merge(1,"A",(o,n)->o+n);
forEach()	Iterate with lambda	map.forEach((k,v)->...);
clone()	Shallow copy	(HashMap<Integer,String>) map.clone();

LinkedHashMap

Method	Description	Example
Preserves insertion order	Iteration order fixed	map.put(1,"Apple");
Access-order mode	Reorders on access	new LinkedHashMap<>(16,0.75f,true);
removeEldestEntry()	Override for LRU cache	protected boolean removeEldestEntry(...)
All HashMap methods	Same as HashMap	map.get(1);

TreeMap

Method	Description	Example
firstKey() / lastKey()	Get smallest/largest key	map.firstKey();
headMap(K) / tailMap(K)	Range views	map.headMap("M");
subMap(K1,K2)	Range subset	map.subMap("A","Z");
higherKey(K) / lowerKey(K)	Next higher/lower key	map.higherKey("Apple");
ceilingKey(K) / floorKey(K)	>= or <= key	map.ceilingKey("Banana");
NavigableMap methods	Extended navigation	map.descendingMap();
All HashMap methods	Similar operations	map.put("A",1);